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21 September 1977
SUPERSEDING
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MILITARY SPECIFICATION

GAUZE, OIL EMULSION IMPREGNATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a sterile, non-adhering fabric of viscose rayon filaments impregnated with a bland water-in-oil emulsion of USP white petrolatum, used as a primary dressing.

1.2 Classification. The impregnated gauze shall be of the following classes and sizes as specified (see 6.2).

Class I -- one dressing per envelope.
Size 1 -- 3 by 3 inches.
Size 2 -- 3 by 8 inches.
Size 3 -- 3 by 16 inches.

Class II - Three dressings per envelope.
Size 2 -- 3 by 8 inches.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, Pa. 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6510

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SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

MILITARY

MIL-L-10547 - Liners, Case and Sheet, Overwrap, Water-vaporproof or Waterproof, Flexible.
MIL-S-36586 - Sterilization Test Strip Set, Bacterial Spore.

STANDARDS

FEDERAL

FED-TEST METHOD STD-191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS STANDARDS (ASTM)
ASTM D88-56 Test for Saybolt Viscosity

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103).

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COLLEGE OF PHYSICIANS OF PHILADELPHIA
Journal of Investigative Determatology Vol 47, Nov 1966, page 393.

(Applications for copies should be addressed to the College of Physicians of Philadelphia, 19 South 22nd St., Philadelphia, PA 19103, ATTN: RML, Photocopy and TWX Section.)

THE U. S. PHARMACOPEIAL CONVENTION, INC. (USP)

The Pharmacopeia of the United States of America

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

3. REQUIREMENTS

3.1 Materials.

3.1.1 Base fabric. The base fabric shall consist of a strip of white warp knit rayon. The fabric shall be knit from bright, 100% viscose rayon yarn. The knitting yarn shall be 150 ± 10 denier, 40 ± 2 filaments. The finished yarn shall have a soft hand with a water soluble spinning finish. The rayon shall conform to the requirements specified in table I, and the finished fabric, to the requirements specified in table II, when tested as specified in 4.5.

TABLE I. Rayon Chemical Requirements.

Characteristic	Requirement
Heavy metal content	15 ppm Max.
Boric acid	0.25% Max.
Total sulfur	50 ppm Max.
Ash	0.5% Max.
Isopropanol extract	0.8% Max.
Total water extract	0.4% Max.
Inorganic water extract	0.2% Max.
pH of water extract	6 Min.
pH of water extract	8.3 Max.

TABLE II. Fabric Requirements.

Characteristic	Requirement
Front end (wales), inch	23 ± 1
Back ends (wales), inch	22 ± 1
Courses, inch	26 ± 2
Weight, grams/lineal yard	6.8 minimum

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3.1.1.1 The design pattern shall be: front bar 1-0, 0-1; back bar 0-0, 3-3. The chain (front bar ends) and lay-in (back bar ends) course and wale stitches shall be evenly spaced in the finished fabric. The right and left selvages shall have non-fraying surfaces. When cut straight or on the bias, the knitted fabric shall not fray or shed loose threads. The fabric shall be free of broken filaments, runs, cuts, oil, dirt, smudges, stains or other manufacturing defects or foregin matter.

3.1.2 Petrolatum emulsion. The petrolatum used in the preparation of the emulsion shall be white petrolatum, a purified unctuous mixture of semisolid hydrocarbons obtained from petroleum, and shall conform to the requirements specified in table III, when tested as specified in 4.3.2.

TABLE III. White Petrolatum, Chemical Requirements.

Characteristic	Requirement
Congealing range	44° - 55° C.
Residue on ignition	0.05% max
Organic acids	Pass
Reaction, alkalinity-acidity	Pass
Fixed oils, fats, resin	Pass
Viscosity, Saybolt	60-80 seconds at 210° F.

3.1.2.1 Petrolatum emulsion type. The petrolatum emulsion used to impregnate the rayon fabric shall be of the water-in-oil type, when tested as specified in 4.5.

3.1.3 Impregnated gauze. The rayon fabric shall be impregnated with the petrolatum emulsion evenly, and uniformly; and shall be non-adhering, porous to drainage, preclusive to maceration and conforming. The emulsion impregnated gauze shall conform to the requirements specified in table IV, and as specified herein. The impregnated gauze strip shall be single-ply.

TABLE IV. Requirements.

Size	Dimensions Unfolded (inches minimum)
1	2 3/4 x 2 3/4
2	2 3/4 x 7 5/8
3 1/	2 3/4 x 15 1/2

1/ Folded in envelope once, approximately in half.

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3.1.3.1 Porosity. The porosity shall be such that 100 ml. of distilled water shall pass through the dressing in not more than 30 seconds, when tested as specified in 4.5.3.

3.1.3.2 Preclusion of maceration. The ability of the impregnated dressing to preclude maceration shall be such as to overcome the surface tension of purified water droplets and permit passage through the dressing in less than 60 seconds, when tested as specified in 4.5.4.

3.1.3.3 Non-adherence. The impregnated dressing shall be non-adherent to wounds, burns or ulcers, and shall not permit clotting to occur within the fabric. This property shall be manifested by non-wicking action of the dressing while suspended in purified water, when tested as specified in 4.5.5.

3.1.3.4 Petrolatum emulsion weight. The weight of the petrolatum emulsion on the impregnated fabric shall be a minimum of 120% of the weight of the untreated fabric (table II) when tested as specified in 4.5.2.

3.2 Sterilization. The impregnated gauze shall be sterile when tested with a biological indicator as specified in 4.5.1.

3.2.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item Identification.
b. National stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.)
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

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When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.2.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number as specified in 5.3.2. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.3 Certificate of dermatological reactions. The contractor shall submit to the procuring activity, a copy of the original certification, and shall indicate that the current lots or batches offered to the Government were composed of the same formula and manufactured identically as the production lot or batch originally tested for dermatological reactions to the impregnated gauze. When tested as specified in 4.5.6 the results of tests shall show a sensitization rate of zero.

3.4 Workmanship. The impregnated dressing shall be free from defects which may affect the appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Inspection lot.

4.2.1 Raw materials. An inspection lot for raw material inspection shall be all material produced under the same conditions and controlled to the degree that uniformity of product within the designated lot can be assured. Lot identification of the raw material by the manufacturer is one method of control, as cited above, which is considered acceptable.

4.2.2 End product. An inspection lot for inspection of end product shall consist of all impregnated gauze of the same type produced under essentially the same conditions and offered for inspection at one time.

4.3 Sampling.

4.3.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table V. The unit of product for sampling purposes shall be one envelope containing the required number of dressings.

TABLE V. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	S-3	1.0
Minor defects	S-3	2.5
For dimensional examination	S-1	2.5

4.3.2 For tests. Except for sterility testing, (see 4.5.1) sampling for tests shall be conducted in accordance with MIL-STD-105 and table VI and as specified herein. The dermatology test shall be run once per lot or a certificate shall be submitted (see 3.3). All other tests shall be run three times per lot and the acceptance number shall be zero. The components listed in table VI shall be tested for characteristics shown therein. The applicable method of testing specified in FED-TEST METHOD STD-191, and as specified herein shall be followed. All test results shall contain the individual values utilized in expressing the final result. For sterility testing the acceptance number shall be zero.

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TABLE VI. Sampling for tests.

Component	Characteristic	Requirement	Test procedure
Base fabric:			
Yarn:	Fiber	3.1.1	1500
	Color	3.1.1	Visual
	Denier	3.1.1	4021
Rayon:			
	Heavy metal content	3.1.1	USP Method II
	Boric acid	3.1.1	4.5.7
	Total sulfur	3.1.1	4.5.8
	Ash	3.1.1	USP
Finish:			
	Isopropanol extract	3.1.1	4.5.9
	Total water extract	3.1.1	4.5.10
	Inorganic water extract	3.1.1	4.5.10
	pH of water extract	3.1.1	2811
Fabric:			
	Front ends (wales)	3.1.1	5070
	Back ends (wales)	3.1.1	5070
	Courses	3.1.1	5070
	Weight	3.1.1	5040
Petrolatum:			
	Congealing range	3.1.2	USP
	Residue on ignition	3.1.2	USP
	Organic acids	3.1.2	USP
	Reaction alkalinty acidity	3.1.2	USP
	Fixed oils, fats with resin	3.1.2	USP
	Viscosity, Saybolt	3.1.2	ASTM D88-56
Petrolatum emulsion:	Type	3.1.2.1	4.5.11
Impregnated gauze:			
	Porosity	3.1.3.1	4.5.3
	Maceration	3.1.3.2	4.5.4
	Non-adherence	3.1.3.3	4.5.5
	Sterility	3.2	4.5.1
	Dermatology	3.3	4.5.6
	Petrolatum emulsion weight	3.1.3.4	4.5.2

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4.4 Examination. The gauze strips shall be examined to determine compliance with all requirements contained in this specification.

4.4.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table VII. Examination shall not be restricted to the classified possible defects listed below.

TABLE VII. Classification of defects.

	Categories and defects
Major:	
101	Right and left selvages, omitted, not non-fraying.
102	Cut edges fray, shed loose threads.
103	Broken filaments, runs, cuts.
104	Not clean - stains, dirt, smudges, foreign matter.
105	Individual envelope not completely sealed.
106	Class II does not contain three dressings per envelope.
107	Labeling incorrect, incomplete, illegible.
Minor:	
201	Size 3 not folded approximately in half.

4.4.2 Dimensional examination. The impregnated gauze shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4.3 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Sterility test. The impregnated gauze shall be tested for sterility in accordance with the method specified in the USP for purified cotton, gauze, surgical dressing and related material or for petrolatum gauze, using inoculated product or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days

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for inoculated product or paper spore strip (carrier). The incubation temperature shall be as indicated in 4.5.1.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for type I sterilization and 30 for type II sterilization including inoculated carrier and product test units.

4.5.1.1 Inoculated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65° C.
Dry heat	B. stearothermophilus, or	55° - 65° C.
	B. subtilis var. niger (or globigii)	30° - 35° C.
Gas	B. subtilis var. niger (or globigii)	30° - 35° C.
Radiation	B. pumilus	30° - 35° C.

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.5.1.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used, shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.5.1 and 4.5.1.1. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.5.1.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.5.1.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.5.1.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the USP.

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4.5.2 Weight of petrolatum emulsion. The weight of the petrolatum emulsion applied to the base fabric (3.1.1) shall be determined by weighing a one yard sample, 3 inches wide of the unimpregnated gauze and a one yard sample, 3 inches wide of the impregnated gauze, prior to packaging. Divide the weight of the impregnated sample by the weight of the unimpregnated sample. Multiply the result by 100 and express as percent pick-up. The percent pick-up shall be calculated as follows:

$$EW, \% = \frac{IR}{R} \times 100$$

Where: EW, % = Weight of emulsion, percent
IR = Impregnated gauze (rayon)
R = Unimpregnated gauze (rayon)

4.5.3 Porosity. The porosity shall be tested by taking a single layer of nonadhering dressing from the envelope, placing it over the rim of a fluted glass funnel, mold-pressed to 60^{0}, top diameter of 50 millimeters (mm.), stem 150 millimeters (mm.). Depress the dressing to the base of the funnel. Simultaneously, pour 100 milliliters (ml.) of purified water into the depressed dressing and start stop watch. Measure the time required for the water to pass through the dressing (3.1.3.1).

4.5.4 Maceration. Place a 4 inch square gauze sponge on a flat surface and place a 3 inch square single ply of nonadhering dressing over the sponge. Place a 7-1/2 inch square of 1/4 inch plexiglass, which has a 1/2 inch hole in the center, over the sample. Carefully place one drop of 0.1% aqueous crystal violet solution on the dressing, in the center of the hole, while simultaneously starting a stop watch. Measure the time to the nearest 1 second required for the dye solution to pass through the nonadhering dressing and be absorbed by the sponge, (see 3.1.3.2).

4.5.5 Non-adherence (non-wicking). Cut a 1 inch x 3 inch strip from a single layer of impregnated dressing. Make the cut so that the long dimension is parallel to the selvage. Suspend the short end of the strip in a beaker cotaining 0.025% aqueous crystal violet solution so that from 2 to 3 mm. of the strip is immersed. Observe the sample during the 5 minute period to determine if any wicking occurs (see 3.1.3.3).

4.5.6 Dermatology test. Dermatological reaction tests shall be performed by the method shown in the Journal of Investigative Dermatology. The test shall be performed on a minimum of twenty-five adults.

4.5.7 Boric acid content.

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4.5.7.1 Apparatus. The test apparatus shall include the following:

a. Culture tubes. 100 by 20 mm., boron-free.

b. Platinum crucible.

c. Spectrophotometer. Equipped with 1 cm. Pyrex cuvettes.

d. Parafilm. Type M.

4.5.7.2 Reagents.

4.5.7.2.1 Carminic acid solution. Dissolve 250 mg. of carminic acid in concentrated sulfuric acid. Dilute to one liter with concentrated sulfuric acid and mix.

4.5.7.2.2 Stock boric acid solution. Weigh approximately 0.15 gm. of sodium tetroborate decahydrate ($NA_2B_4O_7.10\ H_2O$) to the nearest 0.1 mg. into a 500 ml. volumetric flask. Dissolve the borate in purified water, dilute to the mark with purified water and mix. Sodium tetraborate decahydrate contains 64.84% boric acid.

4.5.7.3 Standardization.

Step 1. Pipette 0.0, 5.0, 10.0, 15.0, and 20.0 ml of stock boric acid solution into a series of 100 ml. volumetric flasks.

Step 2. Fill the flasks to the mark with purified water and mix.

Step 3. Pipette 2.0 ml. of one of the standard solutions into a platinum crucible containing 0.10 gm. of lithium carbonate. Repeat this step for each of the remaining solutions.

Step 4. Place the crucibles on the steam bath and evaporate the solution to dryness.

Step 5. Cool the crucibles to room temperature, carefully add 2.0 ml. of the 6 N hydrochloric acid and mix by swirling.

Step 6. Centrifuge the liquid at 3000 rpm. for 2 minutes (min.), or until the supernatant liquid is clear.

Step 7. Pipette 1.0 ml. of the clear supernatant liquid into a boron-free culture tube, add 5.0 ml. of concentrated sulfuric acid, and 5.0 ml. of carminic acid solution. Allow the liberated gases to escape.

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Step 8. Cover the tube with parafilm, shake well and allow to stand for 45 min. at room temperature.

Step 9. Measure the absorbance of the solutions against the reagent blank (0.0 ml. standard) in 1 cm. Pyrex cuvettes at a wavelength of 575 mu (millimicron).

Step 10. Repeat the standardization, using an independently prepared stock boric acid solution.

Step 11. Calculate the regression line of concentration (mcg, microgram, of boric acid per 11.0 ml. of solution) by the method of least squares and from this line calculate the working curve.

4.5.7.4 Sample preparation.

Step 1. Cut the sample into small pieces and weigh approximately 100 mg. to the nearest 0.1 mg. into a platinum crucible.

Step 2. Add 0.10 gm. of lithium carbonate and 2 ml. of purified water. Evaporate the mixture to dryness on a steam bath.

Step 3. Place the crucible on a muffle furnace and heat for 30 min. at 200° C. and for 1.5 hr. at 650° C.

Step 4. Cool the crucible to room temperature, add 2.0 ml. of 6 N hydrochloric acid and mix by swirling.

Step 5. Centrifuge the solution at 3000 rpm. for 2 min., or until the supernatant liquid is clear.

4.5.7.5 Procedure.

Step 1. Pipette 1.0 ml. of the clear supernatant liquid into a boron-free culture tube, add 5.0 ml. of concentrated sulfuric acid, and 5.0 ml. of carminic acid solution.

Step 2. Cover the tube with parafilm, shake well and allow to stand for 45 min. at room temperature.

Step 3. Measure the absorbance of the solution against the reagent blank in 1 cm. Pyrex cuvettes at a wavelength of 575 mu.

4.5.7.6 Calculations. Calculate the concentration of boric acid (mcg. per 11.0 ml.) by means of the working curve. This is the weight (M) of boric acid found.

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$$B = \frac{M}{W \times 1000} \times 100$$

B = percent boric acid content

M = weight of boric acid found - mcg.

W = sample weight - mg.

1000 = conversion factor - mg. to mcg.

4.5.8 Sulfur content.

4.5.8.1 Apparatus. The test apparatus shall include the following:

a. Gas washing bottles. Gas washing bottles equipped with extra coarse sintered glass disc (Corning No. 31760, Size 250 EC or equivalent).

b. Distilling flask. 500 ml. - 3 neck equipped with 24/40 joints (Corning No. 4950 or equivalent).

c. Ground glass stopper. 24/40 joint.

d. Adapter. (Ace Glass No. 5295B or equivalent - cut off the end of the tubing so that there is approximately 2 - 3 mm. between the end of the tube and the bottom of the flask as shown in figure 1.

e. Allihn condenser. 300 mm. equipped with 24/40 joints.

f. Adapter. (Ace Glass No. 5205 or equivalent).

g. Connecting tube. Bend 8 mm. glass tubing to conform to figure 1.

4.5.8.2 Reagents.

4.5.8.2.1 Stannous chloride solution. Dissolve 250 gm. of stannous chloride dihydrate in a mixture of 250. ml. of purified water and 500 ml. of concentrated hydrochloric acid.

4.5.8.2.2 Zinc acetate solution. Dissolve 25 gm. of zinc acetate dihydrate in 500 ml. of purified water. Add 1 ml. of glacial acetic acid, dilute to one liter with purified water and mix.

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4.5.8.2.3 Diamine reagent. Dissolve 1 gm. of p-aminodimethylamine sulfate (Eastman No. 1333 or equivalent) in one liter of 6 N hydrochloric acid.

4.5.8.2.4 Ferric chloride solution. Dissolve 20 gm. of ferric chloride hexahydrate in one liter of 6 N hydrochloric acid.

4.5.8.2.5 Stock sulfur solution. Dry some clear srystals of sodium sulfide enneahydrate ($Na_2S.9\ H_2O$) in a clean cloth and weigh approximately 3.75 gm. to the nearest 10 mg. Dissolve the crystals in recently boiled and cooled purified water and transfer to a 500 ml. volumetric flask. Dilute to the mark with recently boiled and cooled purified water and mix. Discard this solution after 24 hours. Sodium sulfide enneahydrate contains 13.35% sulfur.

4.5.8.2.6 Standard sulfur solution. Pipette 5.0 ml. of stock sulfur solution into a 250 ml. volumetric flask, dilute to the mark with recently boiled and cooled purified water and mix. Discard this solution after 4 hours.

4.5.8.3 Standardization.

Step 1. Add 100 ml. of zinc acetate solution to each of a series of 250 ml. **volumetric flasks.**

Step 2. Measure from a burette 0.0, 1.0, 2.0, 4.0, 6.0, 8.0, 10.0, and 12.0 ml. of standard sulfur solution into the series of flasks and wash down any standard solution adhering to the necks of the flasks with a few ml. of purified water.

Step 3. Swirl to mix and add, while swirling, 50 ml. of diamine reagent and 10 ml. of ferric chloride solution.

Step 4. Dilute to the mark with purified water, mix, and allow to stand 20 min. at room temperature.

Step 5. Measure the absorbance of each of the standard solutions against the reagent blank at a wavelength of 670 **mu. in** 1 cm. cuvettes.

Step 6. Repeat the standardization, using an independently prepared stock solution.

Step 7. Calculate the regression line of absorbance on concentration (mcg. of sulfur/250 ml.) by the method of least squares. From this line calculate the working curve.

4.5.8.4 Sample preparation.

Step 1. Weigh 1 - 2 gm. of sample to the nearest 10 mg. into the 3 neck flask.

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Step 2. Measure 100 ml. of zinc acetate solution into gas washing bottles A, K, and J, figure 1, and assemble the apparatus as shown in the sketch, using short lengths of Tygon tubing for the connections. Add 40 ml. of stannous chloride reagent and 40 ml. of purified water to the flask and replace the stopper.

Step 3. Apply a vacuum to the rubber tubing attached to K and adjust the air flow through the apparatus to a moderate rate (approximately 500 ml. per min.) by means of a screw clamp at L.

Step 4. Boil the contents of the flask at a moderate rate while sweeping the apparatus with the air for 30 min.

Step 5. Shut off the burner and vacuum and disconnect bottles J and K from the apparatus.

Step 6. Disconnect condenser E, adapter F, and connecting Tube I and rinse each into bottle J with a new few mls. of purified water.

4.5.8.5 Procedure.

Step 1. Fill a 50 ml. pipette to the mark with diamine reagent. Raise the sintered disc out of the zinc acetate reagent solution but not out of the bottle and run several ml. of diamine reagent from the pipette through the inlet tube to dissolve any zinc sulfide from the interior of the tube and disc.

Step 2. Add the remainder of the diamine reagent from the pipette directly into the zinc acetate reagent.

Step 3. Immediately add 10 ml. of ferric chloride solution and rinse the interior of the inlet tube and sintered disc into the gas washing bottle with purified water.

Step 4. Transfer the solution quantitatively to a 250 ml. volumetric flask. Dilute to the mark with purified water and mix.

Step 5. Prepare a reagent blank by adding 100 ml. of zinc acetate solution, 50.0 ml. of diamine reagent and 10 ml. of ferric chloride solution, in that order, to a 250 ml. volumetric flask. Dilute to the mark with purified water and mix.

Step 6. Allow the sample and blank to stand for 20 min. at room temperature.

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Step 7. Measure the absorbance of the sample solution against the reagent blank at a wavelength of 670 mu. in a 1 cm. cuvette.

Step 8. Repeat the procedure, using the solution from bottle K.

NOTE: All but the barest trace of hydrogen sulfide is normally absorbed in bottle J. Bottle K should be used at first, but if results indicate that it is superfluous, omit it from the apparatus.

4.5.8.6 Calculations. Calculate the concentration (mcg. of sulfur per 250 ml.) from the working curve. This is the weight (M) of sulfur found.

$$S = \frac{M}{W}$$

S = sulfur content - ppm.

M = weight of sulfur found - mcg.

W = sample weight - gm.

4.5.9 Alcohol extract (Soxhlet method).

4.5.9.1 Apparatus. The test apparatus includes the following:

a. Soxhlet apparatus. The test apparatus consists of:
   1) a flat-bottom round boiling flask with standard taper joints.
   2) a Soxhlet extractor.
   3) a Soxhlet condenser. The apparatus is available in a variety of sizes; the most common being small, medium, and large and measuring 30, 40, and 50 mm. I.D. (extractor), respectively. These sizes normally accomodate 125, 250, and 300 ml. flasks, respectively.

b. Extraction thimbles - paper.

c. Glass wool.

4.5.9.2 Procedure.

Step 1. Weigh approximately 10 gm, of sample to the nearest 10 mg. into a thimble and plug the top of the thimble with glass wool.

Step 2. Place the thimble into a Soxhlet extractor and connect the extractor to the condenser.

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Step 3. Connect a previously dried and tared flask, containing several glass beads and 200 ml. isopropanol to the extractor.

Step 4. Heat the flask with an electric hot plate at such a rate that the solvent cycles at least 6 times per hour. Continue the heating for 2 hrs.

Step 5. Remove the hot plate from under the apparatus, disconnect the flask from the extractor and evaporate the solvent to dryness on a steam bath.

Step 6. Heat the residue in an oven at 105^{0} C. for 1 hr., cool in a desiccator and weigh to the nearest 0.1 mg.

4.5.9.3 Calculations.

$$E = \frac{R}{W} \times 100$$

E = percent

R = weight of residue - gm.

W = sample weight - gm.

4.5.10 Water extract and inorganic extract.

4.5.10.1 Apparatus. The test apparatus includes the following:

a. Soxhlet apparatus. (see 4.5.9.1)

b. Platinum dish.

4.5.10.2 Sample preparation.

Step 1. Air dry the sample used to determine the percent alcohol extract (see 4.5.9).

Step 2. Place this sample in a Soxhlet extractor and connect the extractor to the condenser.

Step 3. Connect the flask containing several glass beads and approximately 225 ml. of purified water.

Step 4. Heat the flask with an electric hot plate at such a rate that the solvent cycles approximately once every 40 min. Continue the heating for 4 cycles.

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Step 5. Disconnect the flask from the extractor prior to a fifth siphoning, leaving most of the water in the extractor.

4.5.10.3 Procedure.

Step 1. Transfer the contents of the flask quantitatively to a tared platinum dish and evaporate the water on a steam bath.

Step 2. Dry the dish and its contents to constant weight in an oven at 105° C. Cool to room temperature in a desiccator and weigh to the nearest 0.1 mg.

Step 3. Place the dish in a cold muffle furnace and ignite at 600° C., until free of carbon.

Step 4. Cool to room temperature in a desiccator and weigh to the nearest 0.1 mg.

4.5.10.4 Calculations.

4.5.10.4.1 Water extract.

$$E = \frac{R}{W} \times 100$$

E = percent water extract

R = weight of water extract - gm.

W = sample weight - gm.

4.5.10.4.2 Inorganic water extract.

$$I = \frac{S}{W} \times 100$$

I = percent inorganic water extract

S = weight of ignited residue - gm.

W = sample weight - gm.

4.5.11 Type of petrolatum emulsion.

4.5.11.1 Electrical conductivity apparatus. The apparatus shall consist of one 1/4 watt neon lamp; one 1/2 watt, 30,000 ohm resistor; one push-button switch; brass bolts and nuts with modified hexagonal head 1/4 x 1/2 inch; (one power cord and plug; one lamp socket; one metal box 4 x 4 x 2 inches;

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one 4 x 4 inch sheet of 1/8 or 1/4 inch thick plastic for mounting parts and top of box. Assemble the parts as shown in figure 2.

4.5.11.2 Standardization.

Step 1. Place a drop of water across the test contacts.

Step 2. Close the circuit by depressing the push-button switch. The neon light should glow brightly.

4.5.11.3 Procedure.

Step 1. Open the sample package and with a spatula, scrape off the coating which adheres to the dressing pad or container backing.

Step 2. Place the coating across the test contacts (alternatively, cut a small piece of the dressing and place it across the test contacts).

Step 3. Close the circuit by depressing the push-button switch. Observe the neon light.

4.5.11.4 Type of emulsion. If the neon light glows faintly or does not glow, the emulsion is a water-in-oil emulsion. If the glow is comparable to the water standard, the emulsion is oil-in-water type.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue.

5.1.1.1 Size 1. One package containing fifty immediate containers of one dressing per immediate container, as specified, constitutes one unit of issue.

5.1.1.2 Size 2. One package containing thirty-six immediate containers of three dressings per immediate container, or one package containing twenty-four immediate containers of one dressing per immediate container, as specified, constitutes one unit of issue.

5.1.1.3 Size 3. One package containing thirty-six immediate containers of one dressing per immediate container, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IX shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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TABLE IX. Packaging quantities.

Size	No. of dressings per immediate container	Unit package	Intermediate package	Exterior container
1	1	1 unit	12 units	48 units
2	3	1 unit	6 units	18 units
2	1	1 unit	Not required	6 units
3	1	1 unit	6 units	24 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Immediate container. The number of dressing specified in table IX for size dressing specified shall be supplied in a commercially available immediate container acceptable to the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.1.3.2 Unit package. Unit package shall be a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-636, type CF, class domestic, or PPP-B-676, except that commercial colors are acceptable. Closure shall be as specified in the appendix of the applicable box specification.

5.1.3.3 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of PPP-B-636.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Fiberboard boxes shall conform to the special requirements as specified in table II of PPP-B-636.

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5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix of PPP-B-636. In addition, waterproofing shall be as specified in PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure of boxes shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Immediate container. In addition to any commercial markings, each container shall be marked in accordance with the requirements of the Federal Food, Drug and Cosmetic Act. Marking shall also include the commerical item identification or the item identification designated as: "GAUZE, OIL EMULSION IMPREGNATED", the size, the number of dressings in an envelope when more than one, manufacturer's name and address, and the legend: "STERILE" AND "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED" or similar statement. Markings may be applied to the container by use of a pressure sensitive label.

5.3.2 Interior packages. Each unit and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable box is fabricated of water-resistant material. Sterility lot (control) number, contract or purchase order number and name of contractor shall be shown. When the contractor is not the manufacturer, the name and address of the manufacturer shall also be shown. In addition, the method of sterilization (G-Gas, R-Radiation, S-Steam, C-Cold, D-Dry Heat Sterilization) shall be shown following the level of packaging and the date of packaging. Example: "A-11/76-G". Final sterilization shall be on or after indicated packaging date.

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5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control number) and the method of sterilization as specified in 5.3.2 shall be shown.

6. NOTES

6.1 Intended use. The oil impregnated gauze covered by this specification is intended for use as a primary, sterile, surgical dressing.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. Class and size required (see 1.2).

6.3 This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification	Class	Size
6510-00-115-0075	GAUZE, OIL EMULSION IMPREGNATED, 3 by 8 inches, 24's	I	2
6510-00-890-1893	GAUZE, OIL EMULSION IMPREGNATED, 3 by 3 inches, 50's	I	1
6510-00-890-1894	GAUZE, OIL EMULSION IMPREGNATED, 3 by 16 inches, 36's	I	3
6510-00-890-1895	GAUZE, OIL EMULSION IMPREGNATED, 3 by 8 inches, 108's	II	2

6.4 Cross reference of classification. This specification includes the following change in classification.

MIL-G-36285B (DM)	MIL-G-36285C
-	Class I
Size 1	Size 1
-	Size 2
Size 3	Size 3
-	Class II
Size 2	Size 2

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity
DoD - MB

Preparing activity:
DSA-DM

Project No. 6510-0618

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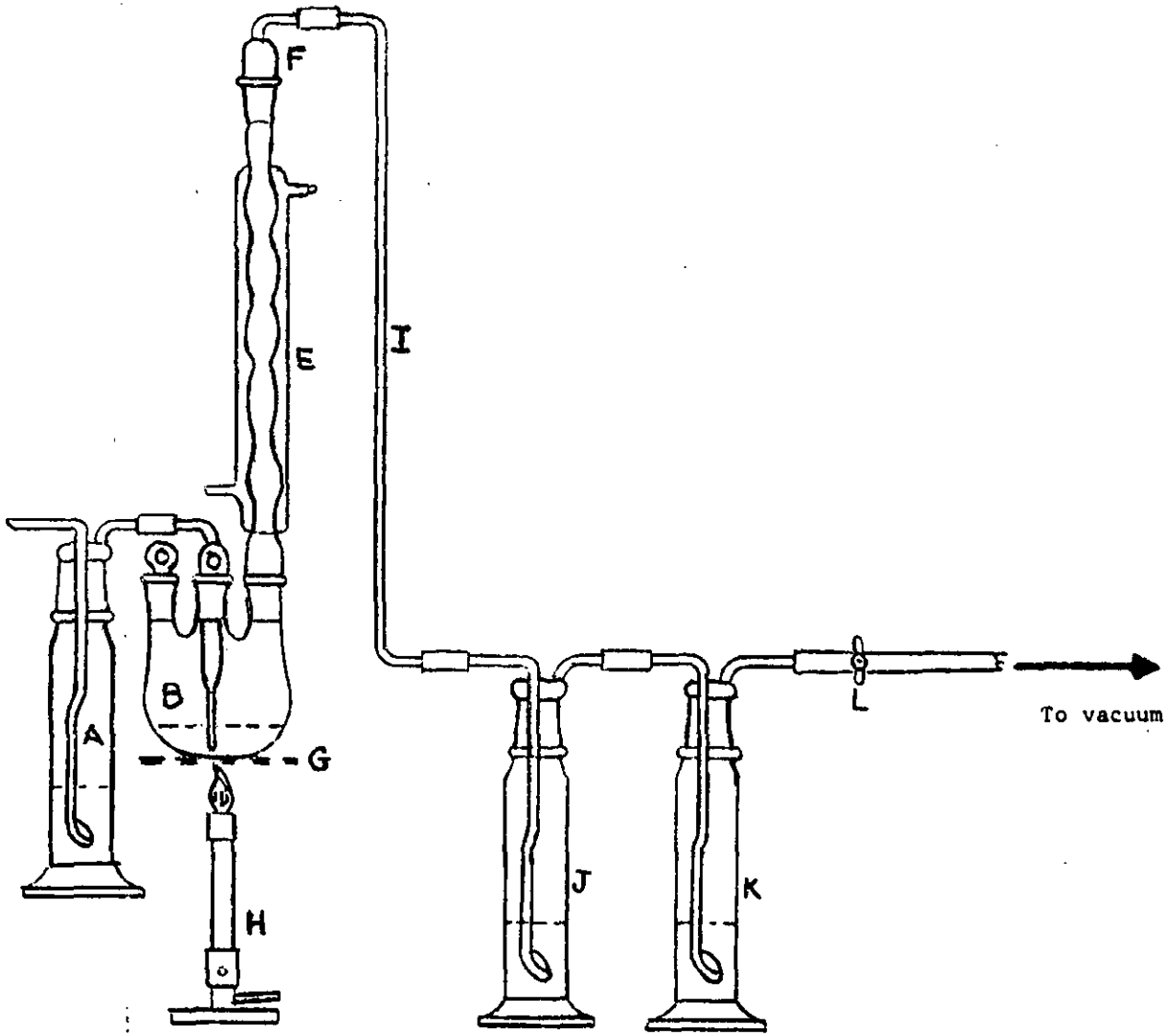


FIGURE I. Apparatus for Sulfur in Rayon Yarns Staple and Tow

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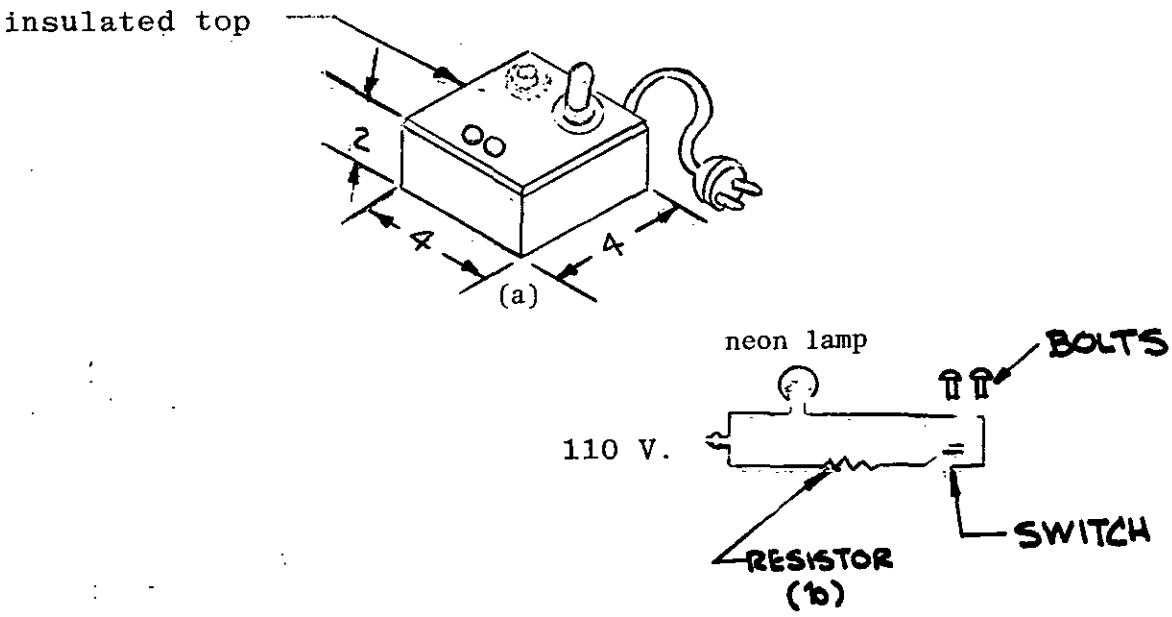


FIGURE 2

(a) View of completed apparatus

(b) Electronic circut

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-G-36285C Gauze, Oil Emulsion Impregnated

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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DATE

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